

$$|J, m\rangle = \sum_{m_1=-J_1}^{J_1} \sum_{m_2=-J_2}^{J_2} |m_1, m_2\rangle \langle m_1, m_2 | J, m \rangle$$

$$1 = \langle J, m | J, m \rangle = \sum_{m_1, m_2} \sum_{m'_1, m'_2} \langle m_1, m_2 | J, m \rangle \langle J, m | m'_1, m'_2 \rangle \underbrace{\langle m'_1, m'_2 | m_1, m_2 \rangle}_{\delta_{m'_1 m_1} \delta_{m'_2 m_2}}$$

$$\Rightarrow 1 = \sum_{m_1, m_2} |\langle m_1, m_2 | J, m \rangle|^2$$

Clebsch Gordon

Clebsch-Gordon - EPQ - 3 ell Eşitsizlikleri

Problem

Bir atomun iki p elektronu $l_1=1$ $l_2=1$ $J=1$ $m=-1$ durumunda iken $m_1=0$ $m_2=-1$ ve $m_1=-1$ $m_2=0$ durumunda bulunma olasılıkları nedir?

$$J=1 \Rightarrow 1, 0, (-1) = m$$

$$J_1 = l_1 \quad J_2 = l_2$$

$$|1, -1\rangle = P_1 |0, -1\rangle + P_2 |-1, 0\rangle \Rightarrow \begin{matrix} m_1=0 & m_2=-1 & |P_1|^2 \\ m_1=-1 & m_2=0 & |P_2|^2 \end{matrix}$$

$$|P_1|^2 + |P_2|^2 = 1$$

$$\Rightarrow J_- |1, -1\rangle = 0 = P_1 J_- |0, -1\rangle + P_2 J_- |-1, 0\rangle$$

$$J_- = J_{1-} + J_{2-}$$

$$\Rightarrow P_1 (J_{1-} + J_{2-}) |0, -1\rangle + P_2 J_- |-1, 0\rangle = 0$$

$l_1=1=l_2$

$$P_1 \hbar \sqrt{2} |-1, -1\rangle + P_2 \hbar \sqrt{2} |-1, -1\rangle = 0$$

$$P_1 + P_2 = 0 \Rightarrow P_1 = -P_2$$

$$|P_1|^2 + |P_2|^2 = 1 \Rightarrow P_1 = \frac{1}{\sqrt{2}} \quad P_2 = -\frac{1}{\sqrt{2}}$$

$$l, m \quad L_- \quad \hbar \sqrt{l(l+1) - m(m-1)}$$

$\sqrt{2-0}$

